Flipkart Machine Design Document

Project Overview

I have used a Flipkart e-commerce dataset that contains information about various products such as product name, category, brand, price, and ratings. The main objective of this project is to perform product and price analysis. This includes analyzing pricing patterns, understanding discount variations, and identifying which products or categories provide better value. The goal is to gain insights into market pricing behavior and help in identifying better deals, which can further support recommendation systems and business decision-making.

Dataset Overview

The dataset contains information about Flipkart products, including details such as product name, category, brand, pricing, and ratings. It is a structured dataset that includes both numerical and categorical features, along with some textual data. This dataset is useful for analyzing product trends, pricing strategies, and customer preferences in an e-commerce environment.

Problem Statement

The rapid growth of e-commerce platforms has led to a large variety of products with different pricing and discount strategies. However, it becomes difficult to understand pricing patterns, identify valuable products, and analyze customer preferences. This project aims to analyze product and pricing data from Flipkart to uncover insights related to pricing distribution, discount trends, brand performance, and customer ratings.

Objectives

- Analyze the distribution of product prices across different categories
- Study discount patterns and calculate discount percentages
- Identify top brands and categories based on product count
- Analyze product ratings and understand customer feedback trends

- Detect outliers in pricing and understand their impact

Tools & Libraries Used

🔹 Programming Language

- Python

🔹 Libraries

- **Pandas** → Data manipulation and preprocessing
- **NumPy** → Numerical operations
- **Matplotlib** → Basic data visualization
- **Seaborn** → Advanced visualizations (heatmaps, distributions)

🔹 Tools

- **Power BI** → Dashboard and visualization
- **Jupyter Notebook** → Development environment

Data Preprocessing

- Loaded dataset using Pandas
- Checked dataset structure (shape, columns, data types)
- Identified missing values using `.isna()` and visualization
- Filled missing values:
  - `retail_price` and `discounted_price` → replaced with median
  - `brand` → filled as "Unknown" where missing
- Removed duplicate records
- Filtered dataset to keep valid `product_rating` values
- Dropped irrelevant columns such as:
  - `uniq_id`, `pid`, `product_url`, `image`, `description`, etc.

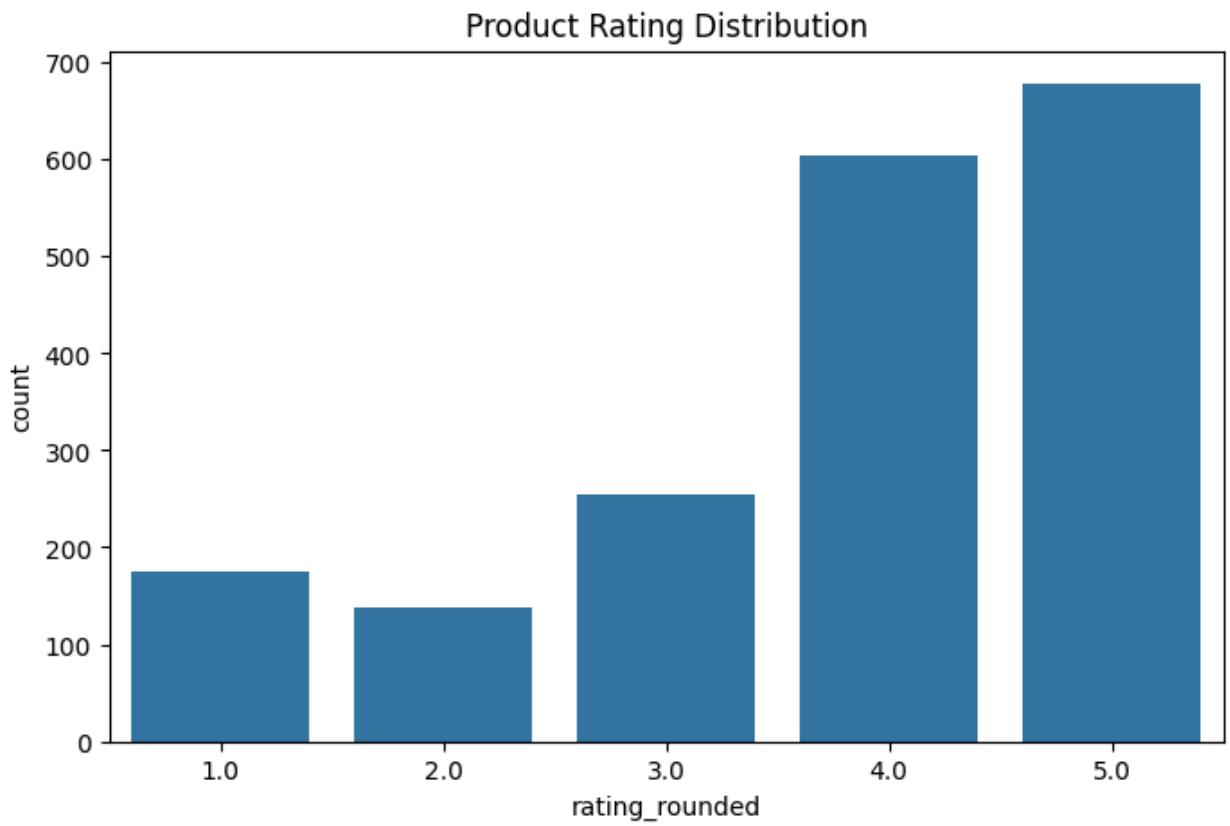
Feature Engineering

- Created new features:
  - **Discount** = retail_price − discounted_price

    - **Discount Percentage** = (discount / retail_price) × 100
- Rounded product ratings for better visualization

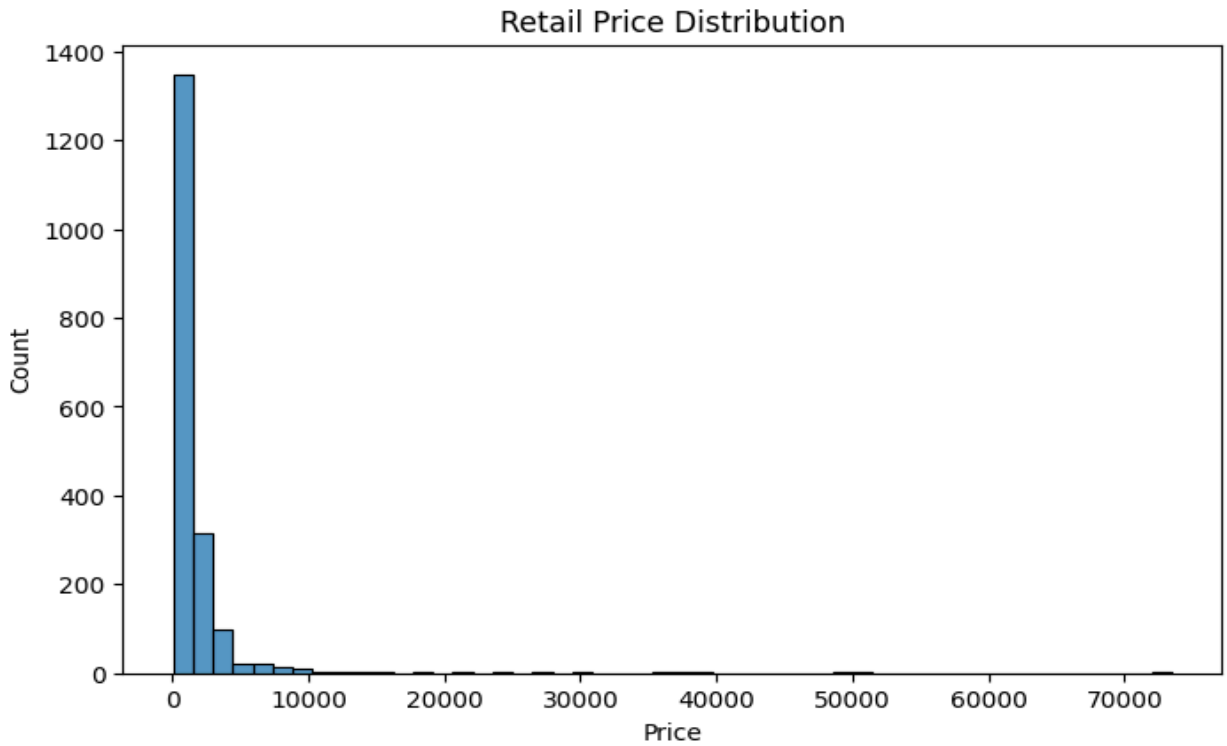
Exploratory Data Analysis (EDA)

Product Ratings Analysis



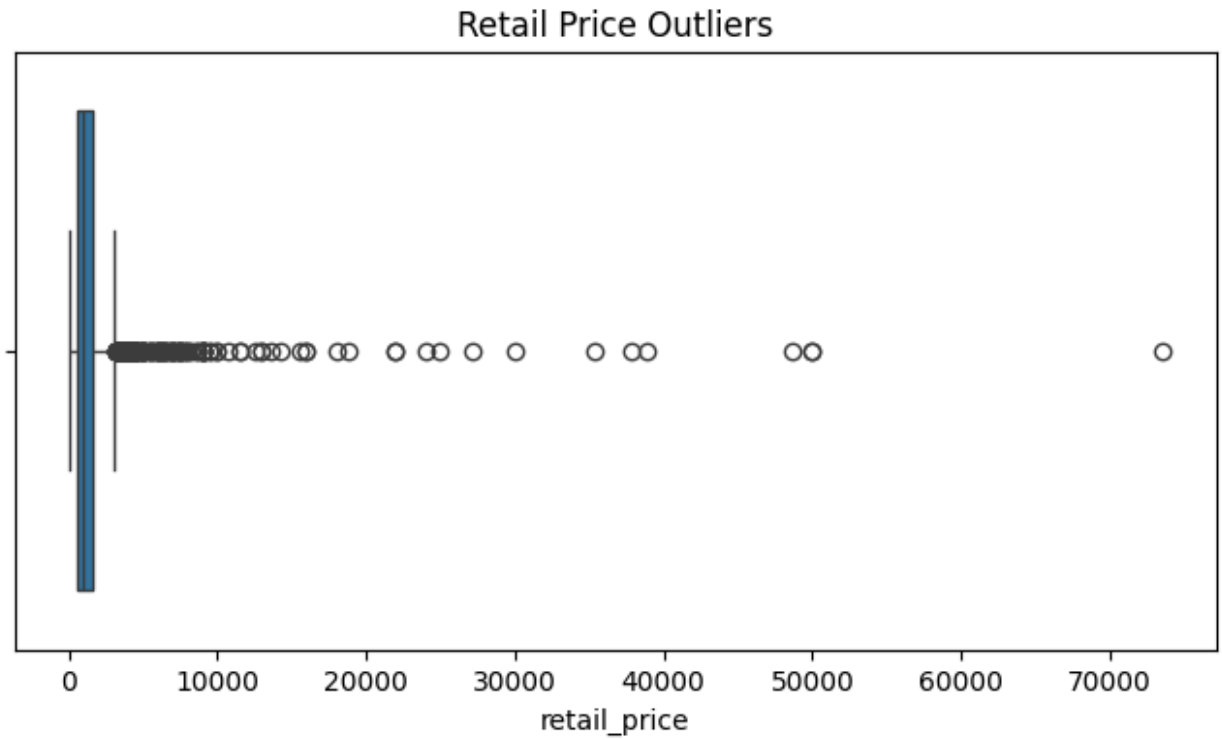
- Most products have ratings between **4 and 5**
- Indicates high customer satisfaction
- Very few products have low ratings

Price Distribution



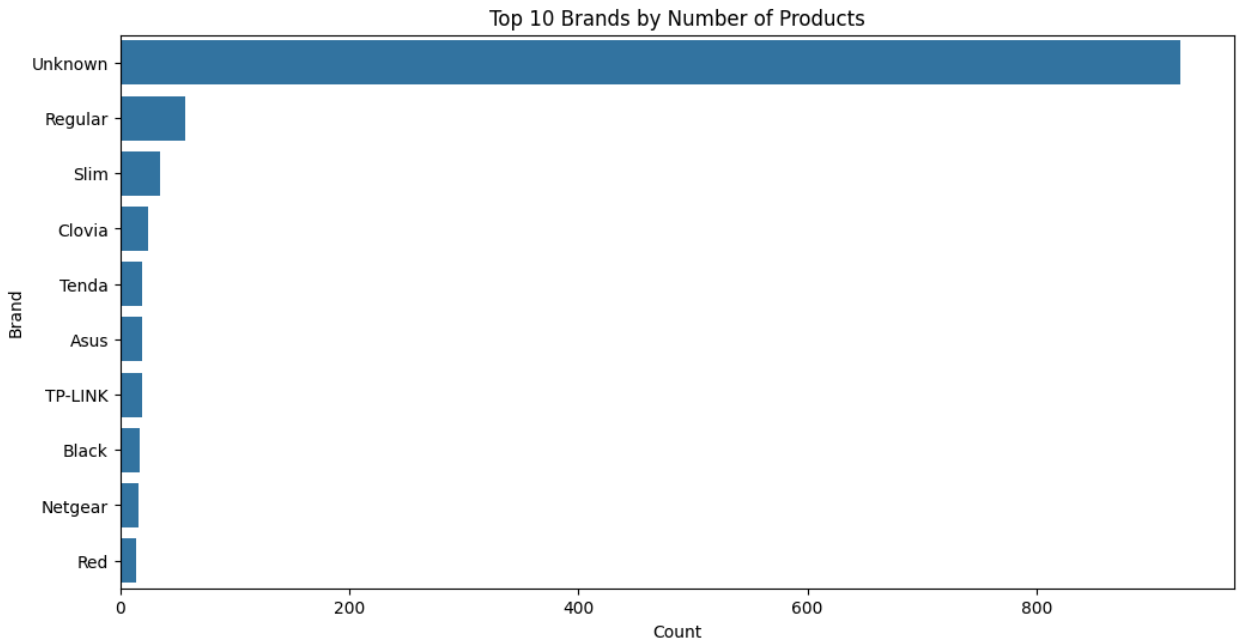
- Majority of products fall in the **₹0–₹2000 range**
- Data is **right-skewed**
- Presence of high-price outliers

Price Outlier



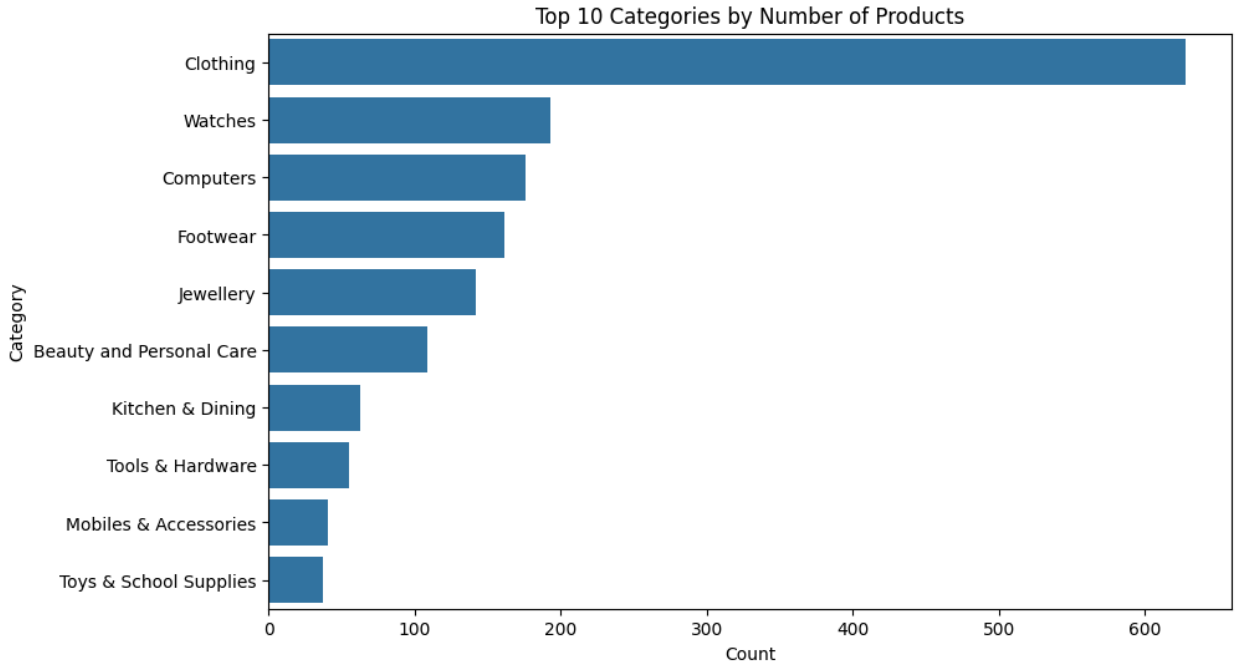
- High-end products exist above ₹50,000
- Majority of products are concentrated in lower price ranges

Top Brands Analysis



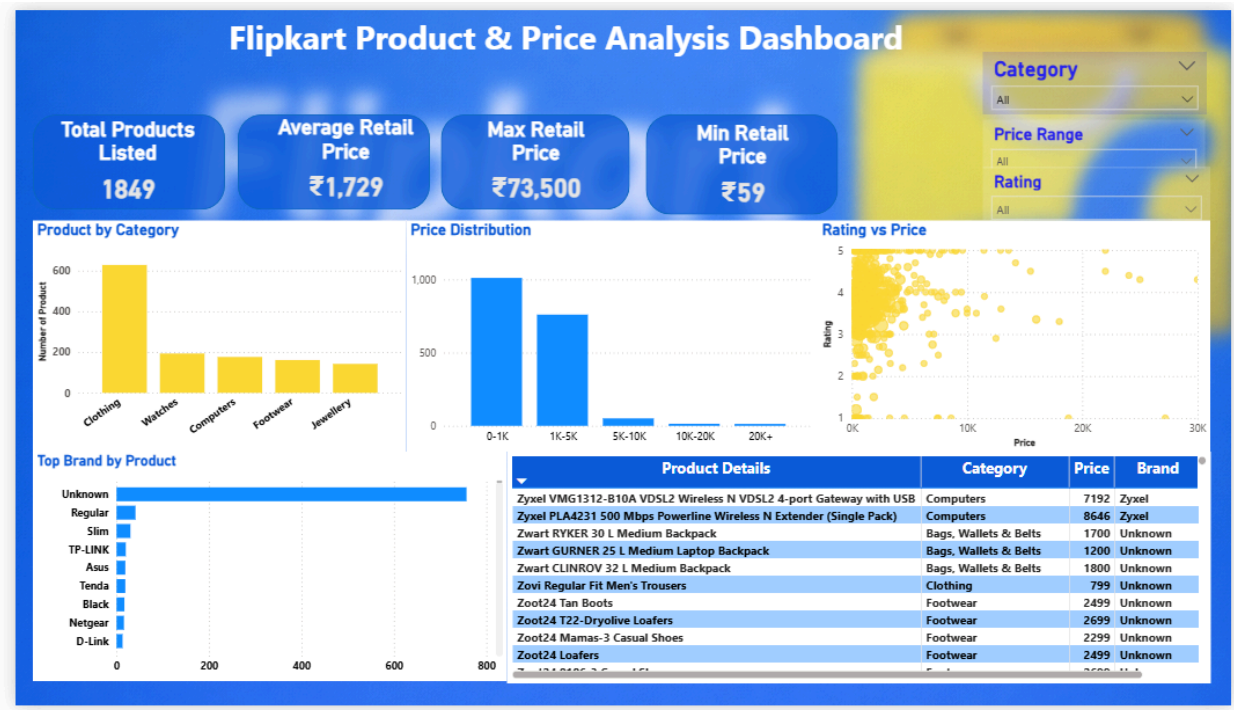
- "Unknown" brand has highest count (due to missing data)
- Other brands like Clovia, Asus, TP-LINK, Netgear appear less frequently

Category Analysis



- Clothing dominates the dataset
- Other categories include Watches, Computers, Footwear
- Some categories have very few products

Dashboard (Power BI)



To better visualize and interpret the insights, an interactive dashboard was created using Power BI.

🔹 Key Metrics:

- Total number of products
- Average retail price
- Maximum and minimum price

🔹 Visualizations:

- Product distribution by category
- Price distribution across ranges
- Rating vs Price (scatter plot)
- Top brands by product count
- Detailed product table

🔹 **Filters:**

- Category
- Rating
- Price range

🔹 **Benefits:**

- Easy understanding of pricing patterns
- Identification of top categories and brands
- Interactive and user-friendly analysis

Key Insights

- **Clothing category dominates** with the highest number of products, while categories like jewellery and footwear have relatively fewer listings.
- Most products fall within the **₹0–₹5,000 price range**, showing a strong focus on affordable items.
- The **average price is ₹1,729**, indicating that the platform mainly targets budget-conscious customers.
- Product ratings are mostly between **4 and 5**, suggesting overall high customer satisfaction.
- There is **no strong relationship between price and rating**, meaning higher-priced products are not necessarily better rated.
- A large number of products belong to **unknown or unbranded sellers**, highlighting a highly competitive and diverse marketplace.

Conclusion

The analysis shows that most products on Flipkart are concentrated in lower price ranges with high customer ratings, indicating affordability and customer satisfaction. Discounts play a significant role in pricing strategies, with noticeable variations across products. Certain categories like clothing dominate the platform, while brand information is often missing or incomplete. The Power BI dashboard enhances the overall analysis by providing interactive and visual insights. These findings can help in understanding market trends, improving pricing strategies, and building better recommendation systems.